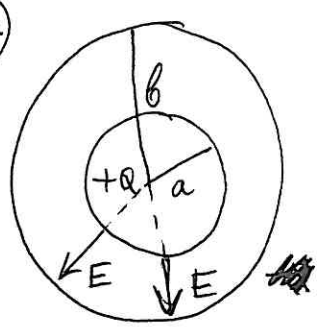
**7.1**

(a)




By symmetry $\vec{E}(\vec{r}) \sim \hat{r}$, is on radial direction

Assume $Q$ on smaller shell

$$\vec{E}(\vec{r}) = \frac{Q}{4\pi\varepsilon_0 r^2}\hat{r}$$

$$V_a - V_b = -\int_b^a \vec{E}(r)\,d\vec{r} =$$

$$= -\frac{Q}{4\pi\varepsilon_0}\int_b^a \frac{dr}{r^2} = \frac{Q}{4\pi\varepsilon_0}\left(\frac{1}{a}-\frac{1}{b}\right)$$

The current $I = \int \vec{J}\cdot d\vec{A} = \int \sigma\vec{E}\cdot d\vec{A} =$

$$= \frac{\sigma Q}{4\pi\varepsilon_0}\int \frac{1}{r^2}\cdot 4\pi r^2 = \frac{\sigma Q}{\varepsilon_0} = \sigma\,\frac{4\pi(V_a - V_b)}{\frac{1}{a}-\frac{1}{b}}$$

(b) $R = \dfrac{V_a - V_b}{I} = \dfrac{1}{4\pi\sigma}\left(\dfrac{1}{a}-\dfrac{1}{b}\right)$

(c) For $b \gg a$ $\quad \dfrac{1}{a}-\dfrac{1}{b} \approx \dfrac{1}{a} \Rightarrow R = \dfrac{1}{4\pi\sigma}\cdot\dfrac{1}{a}$

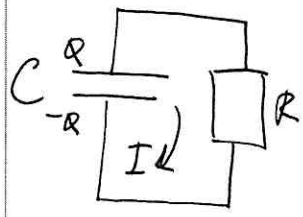
All current leaving (1) enters (2)

$\Rightarrow$ total resistance $R = R_1 + R_2 = 2\cdot\dfrac{1}{4\pi\sigma a} =$

$$= \frac{1}{2\pi a\sigma}$$

where $\sigma$ = conductivity of the medium.

7.2




$$Q_0 = C V_0 .$$

$$V(t) = I(t) R$$

(a) $\frac{dQ}{dt} = -I$ because charge on capacitor is decreasing.

$$\Rightarrow \frac{1}{C} Q = -\frac{dQ}{dt} R \Rightarrow \frac{dQ}{dt} = -\frac{1}{RC} Q$$

$$\Rightarrow Q(t) = Q_0 e^{-t/RC} = C V_0 e^{-t/RC}$$

$$I(t) = -\frac{dQ}{dt} = \frac{C V_0}{RC} e^{-t/RC} = \frac{V_0}{R} e^{-t/RC}$$

(b) $U_0 = \frac{Q_0^2}{2C} = \frac{1}{2} C V_0^2 .$

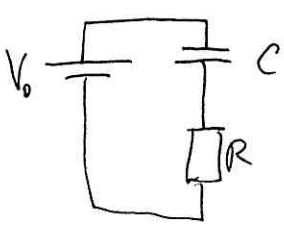
$$P = I^2 R \Rightarrow W = \int_0^\infty I^2 R \, dt = \frac{V_0^2}{R} \int_0^\infty e^{-\frac{2t}{RC}} dt =$$

$$= \frac{V_0^2}{R} \left(-\frac{RC}{2}\right) e^{-2t/RC} \Big|_0^\infty = \frac{V_0^2}{R} \left(-\frac{RC}{2}\right)(-1) =$$

$$= \frac{1}{2} C V_0^2 .$$

$$\Rightarrow W = U_0 .$$

(e)




$$V_0 = \frac{Q}{C} + IR .$$

At $t=0$ charge $Q(t=0)=0 \Rightarrow I = \frac{dQ}{dt}$

$$\Rightarrow V_0 = \frac{Q}{C} + R \frac{dQ}{dt}$$

$$\frac{dQ}{dt} = \frac{CV_0 - Q}{RC} \Rightarrow \frac{dQ}{Q - CV_0} = -\frac{1}{RC} dt \Rightarrow \ln(Q - CV_0) = \frac{-t}{RC} + \alpha$$

constant

$$\Rightarrow Q(t) = CV_0 + e^{\alpha} e^{-t/RC} \quad \text{but } Q(t=0) = 0 \Rightarrow CV_0 + e^{\alpha} = 0$$

$$\Rightarrow Q(t) = CV_0 \left(1 - e^{-t/RC}\right) ; \; I(t) = \frac{dQ}{dt} = \frac{V_0}{R} e^{-t/RC} .$$

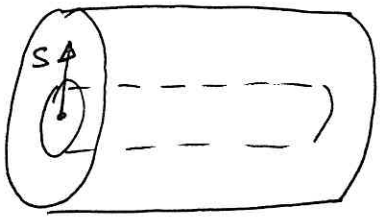
(d) Energy from the battery $= \int_0^\infty V_0 I \, dt =$

$$= \frac{V_0^2}{R} \int_0^\infty e^{-t/RC} \, dt = \frac{V_0^2}{R}(-RC) \underbrace{e^{-t/RC}\Big|_0^\infty}_{-1} = CV_0^2.$$

Since energy on the capacitor $= \frac{Q^2(t=\infty)}{2C} = \frac{(CV_0)^2}{2C} = \frac{1}{2}CV_0^2$

we observe that out of $CV_0^2$ delivered by the battery $\frac{1}{2}CV_0^2$ went to the capacitor, and another $\frac{1}{2}CV_0^2$ was lost on the resistor.

$\boxed{7.4}$




$$\sigma(s) = \frac{k}{s}$$

Total current through cross-section at distance s from the center

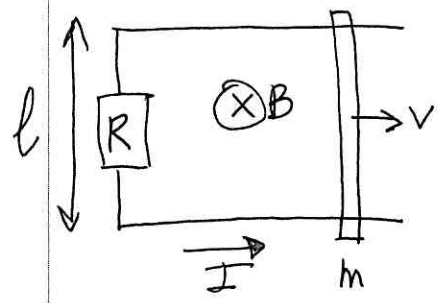
$$\left.\begin{aligned} I &= J \cdot 2\pi s \cdot L \\ J &= \sigma E = \frac{kE}{s} \end{aligned}\right\} I = 2\pi k L E$$

$\Rightarrow E = \frac{I}{2\pi k L}$ in terms of the current.

$$V = -\int_b^a \vec{E} \cdot d\vec{s} = \frac{I}{2\pi k L}(b-a) = IR$$

where $R = \frac{b-a}{2\pi k L}$

7.7




$$\phi = B\ell(x_0 + vt)$$

(a) $\mathcal{E} = -\frac{d\phi}{dt} = -B\ell v$

$I = \frac{B\ell v}{R}$, flows counterclockwise so as to reduce flux $\phi$ which is increasing with time.

(b) $F = I\ell B = \frac{B^2\ell^2 v}{R}$, points to the <u>left</u>. (along $\vec{I} \times \vec{B}$)

(c) $ma = -\frac{B^2\ell^2 v}{R}$, $a = \frac{dv}{dt}$

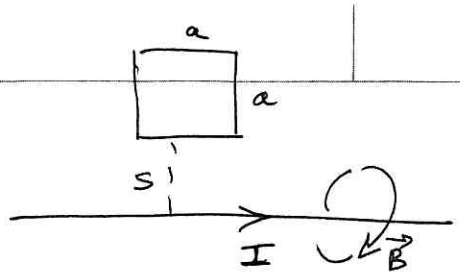
$$\frac{dv}{dt} = -\frac{B^2\ell^2}{mR} v \Rightarrow v(t) = v_0 e^{-\frac{B^2\ell^2}{mR}t}$$

(d) $W = \int_0^\infty I^2 R\,dt = R\left(\frac{B\ell}{R}\right)^2 v_0^2 \int_0^\infty e^{-2ct}\,dt = \frac{B^2\ell^2}{R} v_0^2 \left(\frac{-1}{2c}\right)(-1) =$

$$c = \frac{B^2\ell^2}{mR}$$

$$= \frac{B^2\ell^2}{R} \frac{v_0^2}{2\frac{B^2\ell^2}{mR}} = \frac{1}{2} m v_0^2 \;!$$

**7.8**




$dA = a\,d\rho$

(a) $\vec{B} = \frac{\mu_0 I}{2\pi\rho}\hat{\varphi} \Rightarrow \Phi = \int \frac{\mu_0 I}{2\pi\rho}\,dA = \frac{\mu_0 I a}{2\pi}\int_s^{s+a}\frac{d\rho}{\rho} =$

$$= \frac{\mu_0 I a}{2\pi}\ln\left(\frac{s+a}{s}\right)$$

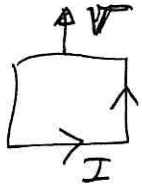
(b) The loop is pulled away $\Rightarrow$ $s(t) = s_0 + vt$

$$\Rightarrow \Phi(t) = \frac{\mu_0 I a}{2\pi}\ln\left(1+\frac{a}{s_0+vt}\right) \xrightarrow[t\to\infty]{} 0$$

$$\mathcal{E} = -\frac{d\Phi}{dt} = -\frac{\mu_0 I a}{2\pi}\cdot\frac{1}{1+\frac{a}{s_0+vt}}\cdot\frac{-av}{(s_0+vt)^2} =$$

$$= \frac{\mu_0 I a^2 v}{2\pi}\,\frac{1}{(s_0+vt)(s_0+vt+a)}$$

Flux (which points out of the page) is diminishing with time $t$ $\Rightarrow$ generated current in the loop flows counterclockwise in order to support it.

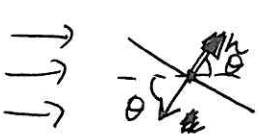



(c) If $\vec{v}$ is along the wire that $s$ does not change

$\Rightarrow \frac{d\Phi}{dt} = 0 \Rightarrow \mathcal{E} = 0.$

**7.10**

top view B




$\Phi = B\cdot a^2\cdot\cos\theta = Ba^2\cos(\omega t+\theta_0) \Rightarrow$

$\theta = \omega t + \theta_0$

$\mathcal{E}(t) = -\frac{d\Phi}{dt} = Ba^2\omega\sin(\omega t+\theta_0).$